

1.Experiment: FIND-S Algorithm

To implement and demonstrate the **FIND-S algorithm** for finding the **most specific hypothesis** using training data.

Python Program

```
import pandas as pd

from google.colab import files

uploaded = files.upload() filename
= list(uploaded.keys())[0] df =
pd.read_csv(filename)

print("Training Data:\n")

print(df)

hypothesis = None for
i in range(len(df)):

    if df.iloc[i, -1] == "Yes": if
        hypothesis is None:
            hypothesis = list(df.iloc[i, :-1])
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != df.iloc[i, j]:
                    hypothesis[j] = "?"
```

```
print("\nFinal Most Specific Hypothesis:")
```

```
print(hypothesis)
```

Output

```
Final Most Specific Hypothesis:  
['?', '?', '?', '?', '?', 'Same', '?', 'Outdoor', '?']
```